

1. State whether the following statement is true or false.

All real numbers are complex numbers.

Choose the correct answer below.

- ☒ True
- ☐ False

2. Determine whether the statement is true or false. If it is false, tell why.

Every pure imaginary number is a complex number.

Choose the correct answer below.

- ☐ True
- ☐ False, no pure imaginary number is a complex number.

3. Determine whether the following statement is true or false.

A number can be both real and complex.

Select the correct choice below.

- ☒ **A.** The given statement is true because any number can be written in the form of a complex number $a + bi$.
- ☐ **B.** The given statement is false because every number cannot be written in the form of a complex number $0 + bi$.
- ☐ **C.** The given statement is false because every number cannot be written in the form of a complex number $a + bi$.
- ☐ **D.** The given statement is true because any number can be written in the form of a complex number $0 + bi$.

4. Identify the number as real, complex, pure imaginary, or nonreal complex. (More than one of these descriptions will apply.)

$2i$

Identify the number $2i$. Select all that apply.

- ☐ **A.** real
- ☐ **B.** pure imaginary
- ☐ **C.** nonreal complex
- ☐ **D.** complex

5. Identify the number as real, complex, pure imaginary, or nonreal complex. (More than one of these descriptions may apply.)

$9 + 7i$

Select all descriptions that may apply.

- ☐ **A.** Real
- ☐ **B.** Pure imaginary
- ☒ **C.** Complex
- ☒ **D.** Nonreal complex

Worked: odd problems

1. $5 = 5 + 0i$ any real $a = a + 0i$

True

3. real $a = a + 0i$, so it is also complex

A

5. $9 + 7i$: $a = 9 \neq 0$, $b = 7 \neq 0$

C. Complex, D. Nonreal complex

6. Identify the number as real, complex, pure imaginary, or nonreal complex. (More than one of these descriptions may apply.)

$$\sqrt{-144}$$

Select all descriptions that may apply.

- ☐ **A.** Real
- ☐ **B.** Complex
- ☐ **C.** Nonreal complex
- ☐ **D.** Pure imaginary

7. Write the number as a product of a real number and i .

$$\sqrt{-19}$$

$$\sqrt{-19} = i\sqrt{19}$$

(Simplify your answer. Type your answer in the form $a + bi$. Type an exact answer, using radicals as needed.)

8. Write the number as the product of a real number and i .

$$\sqrt{-75}$$

$$\sqrt{-75} =$$

(Simplify your answer. Type your answer in the form $a + bi$. Type an exact answer, using radicals as needed.)

9. Multiply.

$$(\sqrt{-28})(\sqrt{-7})$$

$$(\sqrt{-28})(\sqrt{-7}) = -14$$

(Simplify your answer. Type an exact answer, using radicals as needed. Type your answer in the form $a + bi$.)

10. Divide.

$$\frac{\sqrt{-192}}{\sqrt{-64}}$$

$$\frac{\sqrt{-192}}{\sqrt{-64}} =$$

(Simplify your answer. Type your answer in the form $a + bi$. Type an exact answer, using radicals as needed.)

11. Divide.

$$\frac{\sqrt{-28}}{\sqrt{7}}$$

$$\frac{\sqrt{-28}}{\sqrt{7}} = 2i$$

(Simplify your answer. Type your answer in the form $a + bi$. Type an exact answer, using radicals as needed.)

12. Write the number in standard form.

$$\frac{-10 + \sqrt{-40}}{2}$$

$$\frac{-10 + \sqrt{-40}}{2} =$$

(Simplify your answer. Type your answer in the form $a + bi$. Type an exact answer, using radicals as needed.)

7. $\sqrt{-19} = \sqrt{-1} \cdot \sqrt{19} = i\sqrt{19}$

9. $(\sqrt{-28})(\sqrt{-7}) = (2i\sqrt{7})(i\sqrt{7})$
 $= 2i^2(7) = 14i^2 = -14$

11. $\frac{\sqrt{-28}}{\sqrt{7}} = \frac{2i\sqrt{7}}{\sqrt{7}} = 2i$

13. Find the difference between the complex numbers.

$(-2 + 7i) - (-5 + 7i)$

$(-2 + 7i) - (-5 + 7i) =$

(Simplify your answer. Type your answer in the form $a + bi$.)

14. Find the difference between the complex numbers.

$(2 - 7i) - (5 + 4i) - (-3 - 11i)$

$(2 - 7i) - (5 + 4i) - (-3 - 11i) =$

(Simplify your answer. Type your answer in the form $a + bi$.)

15. Simplify.

$(-4 + 2i)(-2 + 6i)$

$(-4 + 2i)(-2 + 6i) =$

(Simplify your answer. Type your answer in the form $a + bi$.)

16. Multiply.

$(7 - 9i)^2$

$(7 - 9i)^2 =$

(Simplify your answer. Type your answer in the form $a + bi$.)

17. Find the product and write the result in standard form, $a + bi$.

$(9 + i)(9 - i)$

$(9 + i)(9 - i) =$

(Simplify your answer. Type your answer in the form $a + bi$.)

18. Find the product.

$(-4 - 3i)(-4 + 3i)$

$(-4 - 3i)(-4 + 3i) =$ (Type your answer in the form $a + bi$.)

19. Find the product and write the result in standard form.

$i(6 + 5i)(6 - 5i)$

$i(6 + 5i)(6 - 5i) =$

(Type an integer or a simplified fraction. Type your answer in the form $a + bi$.)

20. Divide.

$$\frac{1 + 7i}{1 + 8i}$$

$\frac{1 + 7i}{1 + 8i} =$

(Simplify your answer. Use integers or fractions for any numbers in the expression. Type your answer in the form $a + bi$.)

Worked: odd problems

13. $-2 + 7i + 5 - 7i =$

15. $8 - 24i - 4i + 12i^2$
 $= 8 - 28i - 12 =$

17. $81 - 9i + 9i - i^2 = 81 + 1 =$

19. $i(36 - 25i^2) = i(36 + 25) =$

21. Find the quotient. Write the answer in standard form $a + bi$.

$$\frac{2}{-i}$$

$$\frac{2}{-i} = 2i$$

(Simplify your answer. Type your answer in the form $a + bi$.)

22. Solve the quadratic equation and express all nonreal complex solutions in terms of i .

$$x^2 + 20 = 0$$

The solution set is $\{ \quad \}$.

(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Express complex numbers in terms of i . Use a comma to separate answers as needed.)

23. Solve $7x^2 + 3 = -4x$ over the set of complex numbers.

The solution set is $\left\{ -\frac{2}{7} + (\sqrt{17}/7)i, -\frac{2}{7} - (\sqrt{17}/7)i \right\}$

(Use integers or fractions for any numbers in the expression. Express complex numbers in terms of i . Use a comma to separate answers as needed.)

24. Simplify.

¹⁹_{*l*}

$$i^{19} = \boxed{}$$

25. Find the power of i .

$i-13$

$$i^{-13} = \boxed{-i}$$

(Simplify your answer. Type your answer in the form $a + bi$.)

26. Simplify the power of i .

$$\frac{1}{l^{-15}}$$

$$\frac{1}{i^{-15}} = \boxed{}$$

(Simplify your answer. Type your answer in the form $a + bi$.)

Worked: odd problems

21. $\frac{2}{-i} \cdot \frac{i}{i} = \frac{2i}{-i^2} = \frac{2i}{1} = \boxed{2i}$

23. $7x^2 + 4x + 3 = 0$

$$x = \frac{-4 \pm \sqrt{16 - 4(7)(3)}}{2(7)} = \frac{-4 \pm \sqrt{-68}}{14}$$

$$= \frac{-4 \pm 2i\sqrt{17}}{14}$$

$$x = \boxed{-\frac{2}{7} \pm \frac{\sqrt{17}}{7}i}$$

25. $i^{-13} = \frac{1}{i^{13}}, \quad 13 \div 4 = 3 \text{ R } 1, \quad i^{13} = i$

$$\frac{1}{i} \cdot \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1} = \boxed{-i}$$

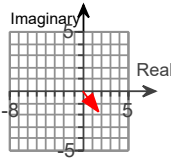
27. Graph the complex number as a vector in the complex plane.

$-\sqrt{3} + \sqrt{3}i$

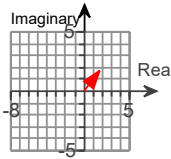
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Choose the correct graph below.

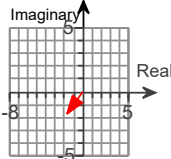
☐ A.



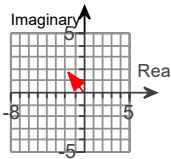
☐ B.



☐ C.



☒ D.

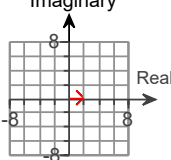


28. Graph the complex number as a vector in the complex plane.

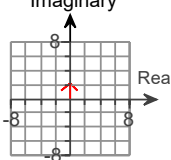
$-2i$

Choose the correct graph of the vector.

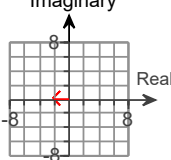
☐ A.



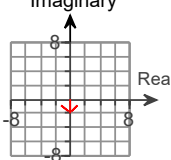
☐ B.



☐ C.



☐ D.



29. Write the complex number in rectangular form.

$18(\cos(45^\circ) + i \sin(45^\circ))$

Choose the correct answer below.

☐ A. $\sqrt{2} - i\sqrt{2}$

☒ B. $9\sqrt{2} + 9i\sqrt{2}$

☐ C. $9\sqrt{2} - 9i\sqrt{2}$

☐ D. $\sqrt{2} + i\sqrt{2}$

30. Write the complex number in rectangular form.

$9 \text{ cis } 120^\circ$

The complex number is .

(Simplify your answer, including any radicals. Type your answer in the form $a + bi$. Use integers or fractions for any numbers in the expression.)

Worked: odd problems

27. $(-\sqrt{3}, \sqrt{3}) \approx (-1.7, 1.7)$, QII

D

29. $18\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) = 9\sqrt{2} + 9i\sqrt{2}$

B

31. Write the complex number in rectangular form.

$20(\cos(-45^\circ) + i \sin(-45^\circ))$

The complex number is $10\sqrt{2} - 10i\sqrt{2}$.
(Simplify your answer, including any radicals. Type your answer in the form a + bi. Use integers or fractions for any numbers in the expression.)

32. Write the complex number in trigonometric form $r(\cos \theta + i \sin \theta)$, with θ in the interval $[0^\circ, 360^\circ)$.

$4 + 4i$

$4 + 4i =$ $(\cos$ $^\circ + i \sin$ $^\circ)$
(Type the value for r as an exact answer, using radicals as needed. Type the value for θ as an integer or decimal rounded to the nearest tenth as needed.)

33. Write the complex number in trigonometric form $r(\cos \theta + i \sin \theta)$, with θ in the interval $[0^\circ, 360^\circ)$.

$5i$

$5i =$ $(\cos$ $^\circ + i \sin$ $^\circ)$

34. Is $z = -0.16i$ in the Julia set?

Choose the correct answer below.

- ☐ Yes
☐ No

35. Find the following product, and write the product in rectangular form.

$[2(\cos 240^\circ + i \sin 240^\circ)][7(\cos 165^\circ + i \sin 165^\circ)]$

$[2(\cos 240^\circ + i \sin 240^\circ)][7(\cos 165^\circ + i \sin 165^\circ)] =$
(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in the form a + bi.)

36. Find the following product, and write the product in rectangular form.

$(\sqrt{5} \operatorname{cis} 60^\circ)(\sqrt{5} \operatorname{cis} 30^\circ)$

$(\sqrt{5} \operatorname{cis} 60^\circ)(\sqrt{5} \operatorname{cis} 30^\circ) =$
(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in the form a + bi.)

37. Find the quotient, and write it in rectangular form.

$\frac{18(\cos 310^\circ + i \sin 310^\circ)}{3(\cos 130^\circ + i \sin 130^\circ)}$

$\frac{18(\cos 310^\circ + i \sin 310^\circ)}{3(\cos 130^\circ + i \sin 130^\circ)} =$
(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in the form a + bi.)

Worked: odd problems

31. $\cos(-45^\circ) = \frac{\sqrt{2}}{2}, \sin(-45^\circ) = -\frac{\sqrt{2}}{2}$
 $20\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) =$

33. $r = \sqrt{0^2 + 5^2} = 5, \theta = 90^\circ$ (positive imaginary axis)

35. $2 \cdot 7 \operatorname{cis}(240^\circ + 165^\circ) = 14 \operatorname{cis} 405^\circ$
 $= 14 \operatorname{cis} 45^\circ = 14\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)$
 $=$

37. $\frac{18}{3} \operatorname{cis}(310^\circ - 130^\circ) = 6 \operatorname{cis} 180^\circ = 6(-1 + 0i) =$

38. Write the quotient in rectangular form.

$$\frac{2 \operatorname{cis} 170^{\circ}}{10 \operatorname{cis} 35^{\circ}}$$

$$\frac{2 \operatorname{cis} 170^{\circ}}{10 \operatorname{cis} 35^{\circ}} = \boxed{}$$

(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in the form $a + bi$.)

39. Find the quotient and write it in rectangular form.

$$\frac{7\sqrt{21} + 7i\sqrt{7}}{\sqrt{7} - i\sqrt{21}}$$

$$\frac{7\sqrt{21} + 7i\sqrt{7}}{\sqrt{7} - i\sqrt{21}} = \boxed{7i}$$

(Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in the form $a + bi$.)

40. Raise the number to the given power and write the answer in rectangular form.

$$[\sqrt{6}(\operatorname{cis} 85^{\circ})]^6$$

$$[\sqrt{6}(\operatorname{cis} 85^{\circ})]^6 = \boxed{}$$

(Type an exact answer, using radicals as needed.)

41. Find the given power.

$$(3 + 3i)^7$$

The answer is $\boxed{17496 - 17496i}$

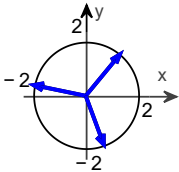
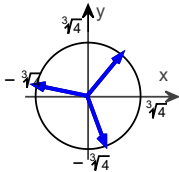
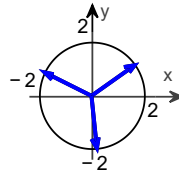
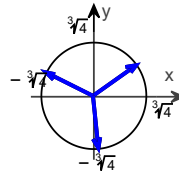
(Type your answer in the form $a + bi$.)

42. Find and graph the cube roots of $-2\sqrt{3} + 2i$.

Choose the correct cube roots of $-2\sqrt{3} + 2i$ below.

- ☐ A. $2(\cos 50^{\circ} + i \sin 50^{\circ})$, $2(\cos 170^{\circ} + i \sin 170^{\circ})$, $2(\cos 290^{\circ} + i \sin 290^{\circ})$
- ☐ B. $\sqrt[3]{4}(\cos 50^{\circ} + i \sin 50^{\circ})$, $\sqrt[3]{4}(\cos 170^{\circ} + i \sin 170^{\circ})$, $\sqrt[3]{4}(\cos 290^{\circ} + i \sin 290^{\circ})$
- ☐ C. $\sqrt[3]{4}(\cos 35^{\circ} + i \sin 35^{\circ})$, $\sqrt[3]{4}(\cos 155^{\circ} + i \sin 155^{\circ})$, $\sqrt[3]{4}(\cos 275^{\circ} + i \sin 275^{\circ})$
- ☐ D. $2(\cos 35^{\circ} + i \sin 35^{\circ})$, $2(\cos 155^{\circ} + i \sin 155^{\circ})$, $2(\cos 275^{\circ} + i \sin 275^{\circ})$

Choose the correct graph of the cube roots of $-2\sqrt{3} + 2i$ below.

- ☐ A. 
- ☐ B. 
- ☐ C. 
- ☐ D. 

39. $r_1 = \sqrt{(7\sqrt{21})^2 + (7\sqrt{7})^2} = \sqrt{1029 + 343} = \sqrt{1372} = 14\sqrt{7}$
 $\theta_1 = \tan^{-1} \frac{7\sqrt{7}}{7\sqrt{21}} = \tan^{-1} \frac{1}{\sqrt{3}} = 30^{\circ}$
 $r_2 = \sqrt{7 + 21} = \sqrt{28} = 2\sqrt{7}$
 $\theta_2 = \tan^{-1} \left(-\frac{\sqrt{21}}{\sqrt{7}} \right) = \tan^{-1}(-\sqrt{3}) = -60^{\circ} \rightarrow 300^{\circ}$
 $\frac{14\sqrt{7} \operatorname{cis} 30^{\circ}}{2\sqrt{7} \operatorname{cis} 300^{\circ}} = 7 \operatorname{cis}(-270^{\circ}) = 7 \operatorname{cis} 90^{\circ} = \boxed{7i}$

41. $r = \sqrt{9 + 9} = 3\sqrt{2}$, $\theta = \tan^{-1}(1) = 45^{\circ}$
 $(3\sqrt{2})^7 \operatorname{cis}(7 \cdot 45^{\circ}) = 3^7(\sqrt{2})^7 \operatorname{cis} 315^{\circ}$
 $= 2187 \cdot 8\sqrt{2} \operatorname{cis} 315^{\circ} = 17496\sqrt{2} \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right)$
 $= \boxed{17496 - 17496i}$

43. Find all complex number solutions of the equation.

$x^3 - 8i = 0$

Choose the correct solution set below.

- ☐ A. $\{8(\cos 30^\circ + i \sin 30^\circ), 8(\cos 150^\circ + i \sin 150^\circ), 8(\cos 270^\circ + i \sin 270^\circ)\}$
- ☐ B. $\{8(\cos 90^\circ + i \sin 90^\circ), 8(\cos 210^\circ + i \sin 210^\circ), 8(\cos 330^\circ + i \sin 330^\circ)\}$
- ☒ C. $\{2(\cos 30^\circ + i \sin 30^\circ), 2(\cos 150^\circ + i \sin 150^\circ), 2(\cos 270^\circ + i \sin 270^\circ)\}$
- ☐ D. $\{2(\cos 90^\circ + i \sin 90^\circ), 2(\cos 210^\circ + i \sin 210^\circ), 2(\cos 330^\circ + i \sin 330^\circ)\}$

44. Find all complex solutions.

$x^4 + 8 = 0$

Choose the correct solutions below.

- ☐ A. $\{8 \text{ cis } 0^\circ, 8 \text{ cis } 90^\circ, 8 \text{ cis } 180^\circ, 8 \text{ cis } 270^\circ\}$
- ☐ B. $\{\sqrt[4]{8} \text{ cis } 0^\circ, \sqrt[4]{8} \text{ cis } 90^\circ, \sqrt[4]{8} \text{ cis } 180^\circ, \sqrt[4]{8} \text{ cis } 270^\circ\}$
- ☐ C. $\{\sqrt[4]{8} \text{ cis } 45^\circ, \sqrt[4]{8} \text{ cis } 135^\circ, \sqrt[4]{8} \text{ cis } 225^\circ, \sqrt[4]{8} \text{ cis } 315^\circ\}$
- ☐ D. $\{8 \text{ cis } 45^\circ, 8 \text{ cis } 135^\circ, 8 \text{ cis } 225^\circ, 8 \text{ cis } 315^\circ\}$

45. Find all complex number solutions of the equation.

$x^3 - (32 + 32i\sqrt{3}) = 0$

Choose the correct solution set below.

- ☒ A. $\{4(\cos 20^\circ + i \sin 20^\circ), 4(\cos 140^\circ + i \sin 140^\circ), 4(\cos 260^\circ + i \sin 260^\circ)\}$
- ☐ B. $\{64(\cos 20^\circ + i \sin 20^\circ), 64(\cos 140^\circ + i \sin 140^\circ), 64(\cos 260^\circ + i \sin 260^\circ)\}$
- ☐ C. $\{4(\cos 60^\circ + i \sin 60^\circ)\}$
- ☐ D. $\{64(\cos 60^\circ + i \sin 60^\circ)\}$

Worked: odd problems

43. $x^3 = 8i = 8 \text{ cis } 90^\circ, \sqrt[3]{8} = 2$
 $\alpha = \frac{90^\circ + 360^\circ k}{3} = 30^\circ + 120^\circ k$
 $k = 0, 1, 2 : 30^\circ, 150^\circ, 270^\circ$

C

45. $x^3 = 32 + 32i\sqrt{3}$
 $r = \sqrt{32^2 + (32\sqrt{3})^2} = \sqrt{1024 + 3072} = \sqrt{4096} = 64$
 $\theta = \tan^{-1}(\sqrt{3}) = 60^\circ, \sqrt[3]{64} = 4$
 $\alpha = \frac{60^\circ + 360^\circ k}{3} = 20^\circ + 120^\circ k : 20^\circ, 140^\circ, 260^\circ$

A

46. The fractal called the Mandelbrot set is shown in the figure. To determine if a complex number $z = a + bi$ is in this set, perform the following sequence of calculations. Repeatedly compute the following.

$$z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$$

The complex number z does not belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set and the point (a,b) should be shaded in the graph. Determine whether or not the following numbers belong to the Mandelbrot set.



(a) $z = 0 + 0i$

- ☐ A. This number does not belong to the Mandelbrot set.
- ☐ B. This number belongs to the Mandelbrot set.

(b) $z = 1 + 1i$

- ☐ A. This number does not belong to the Mandelbrot set.
- ☐ B. This number belongs to the Mandelbrot set.

(c) $z = 0 - 0.1i$

- ☐ A. This number does not belong to the Mandelbrot set.
- ☐ B. This number belongs to the Mandelbrot set.

47. For the following point given in polar coordinates, state the quadrant in which the point lies if it is graphed in a rectangular coordinate system.

$(2, 120^\circ)$

Choose the correct quadrant in which $(2, 120^\circ)$ lies.

- ☐ I
- ☐ IV
- ☒ II
- ☐ III

48. For the point $(-1, 30^\circ)$ in polar coordinates, (a) plot the point, (b) give two other pairs of polar coordinates for the point, and (c) give the rectangular coordinates for the point.

a) Determine the correct graph below.

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

b) Determine the other two pairs of polar coordinates of $(-1, 30^\circ)$. Select all that apply.

- ☐ A. $(1, 30^\circ)$
- ☐ B. $(1, 210^\circ)$
- ☐ C. $(1, 390^\circ)$
- ☐ D. $(-1, 390^\circ)$

c) The rectangular coordinates of $(-1, 30^\circ)$ are .

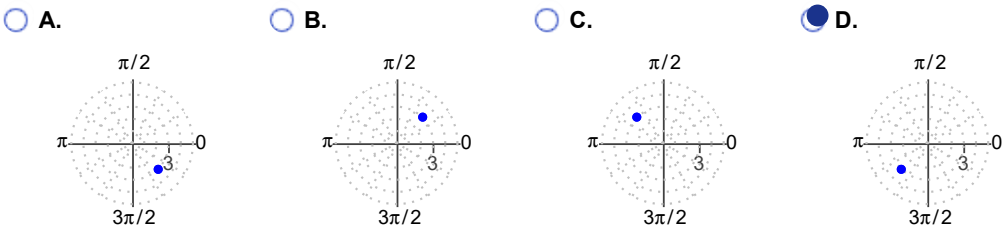
(Type an ordered pair. Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression.)

47. $r = 2 > 0, \theta = 120^\circ$ is in QII

II

49. For the point $\left(3, \frac{5\pi}{4}\right)$ in polar coordinates, **(a)** plot the point, **(b)** give two other pairs of polar coordinates for the point, and **(c)** give the rectangular coordinates of the point.

(a) Choose the graph that represents the point $\left(3, \frac{5\pi}{4}\right)$.



(b) What are the coordinates of the point for which $r > 0$ and $2\pi < \theta < 4\pi$?

(Type an ordered pair. Simplify your answer. Type an exact answer using π as needed. Use integers or fractions for any numbers in the expression.)

What are the coordinates of the point for which $r < 0$ and $0 < \theta < 2\pi$?

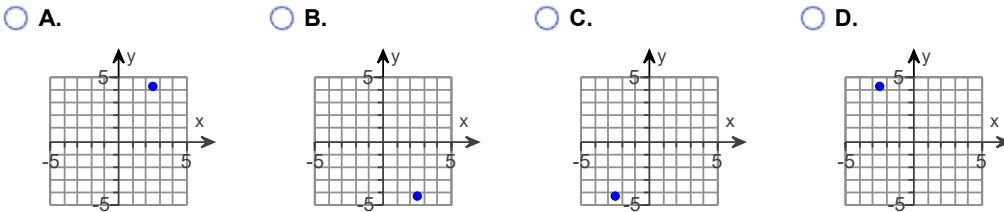
(Type an ordered pair. Simplify your answer. Type an exact answer using π as needed. Use integers or fractions for any numbers in the expression.)

(c) In rectangular coordinates, the point $\left(3, \frac{5\pi}{4}\right)$ is

(Type an ordered pair. Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression.)

50. For the rectangular coordinates $\left(\frac{5}{2}, -\frac{5\sqrt{3}}{2}\right)$, **(a)** plot the point and **(b)** give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$.

(a) Choose the correct plot of $\left(\frac{5}{2}, -\frac{5\sqrt{3}}{2}\right)$ below.



(b) Give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$.

(Type an ordered pair. Simplify your answer, including any radicals. Use integers or fractions for any numbers in the expression. Type your answer in degrees. Do not include the degree symbol in your answer. Use a comma to separate answers as needed.)

49. (a) $\frac{5\pi}{4}$ is QIII, $r > 0$:

(b) $\frac{5\pi}{4} + 2\pi = \frac{13\pi}{4}$:

$\frac{5\pi}{4} - \pi = \frac{\pi}{4}$:

(c) $x = 3 \cos \frac{5\pi}{4} = 3 \left(-\frac{\sqrt{2}}{2}\right) = -\frac{3\sqrt{2}}{2}$

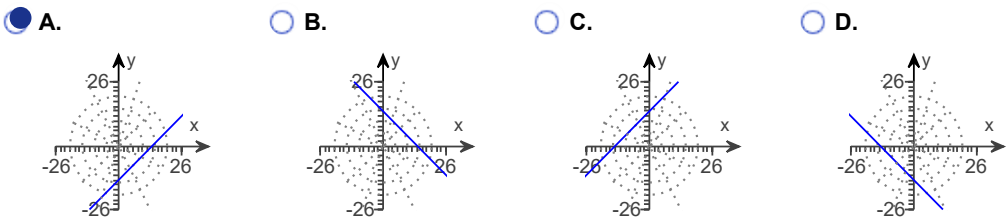
$y = 3 \sin \frac{5\pi}{4} = 3 \left(-\frac{\sqrt{2}}{2}\right) = -\frac{3\sqrt{2}}{2}$

51. For the following rectangular equation, give the equivalent polar equation and sketch its graph.

$x - y = 14$

The polar form of the rectangular equation is $r = 14/(\cos \theta - \sin \theta)$

Graph the polar equation. Choose the correct graph below.



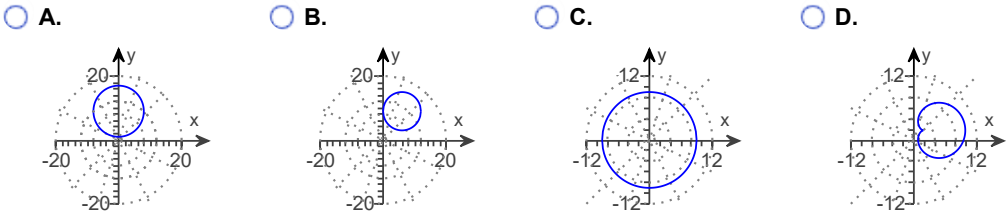
52. For the following rectangular equation, give the equivalent polar equation and sketch its graph.

$x^2 + y^2 = 81$

The polar form of the rectangular equation is

(Use a comma to separate answers as needed. Type each answer only once.)

Graph the polar equation. Choose the correct graph below.

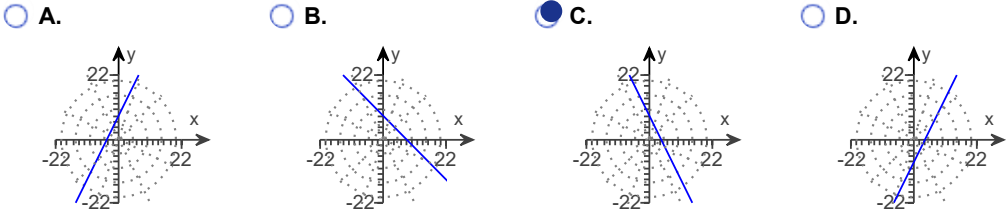


53. For the following rectangular equation, give the equivalent polar equation and sketch its graph.

$2x + y = 8$

The polar form of the rectangular equation is $r = 8/(2 \cos \theta + \sin \theta)$

Graph the polar equation. Choose the correct graph below.



51. $r \cos \theta - r \sin \theta = 14$

$r(\cos \theta - \sin \theta) = 14$

$r = \frac{14}{\cos \theta - \sin \theta}$

$y = x - 14$: slope 1, y -int -14 : **A**

53. $2r \cos \theta + r \sin \theta = 8$

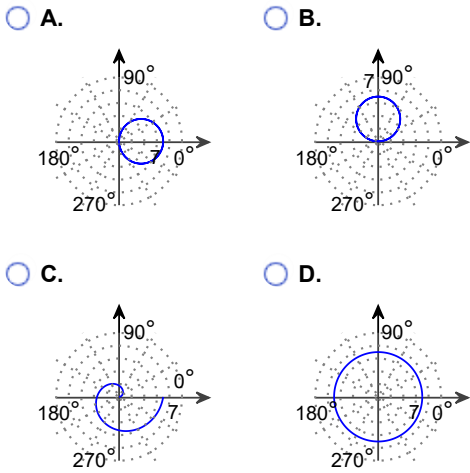
$r = \frac{8}{2 \cos \theta + \sin \theta}$

$y = -2x + 8$: slope -2 , y -int 8 : **C**

54. Find the polar graph of the equation.

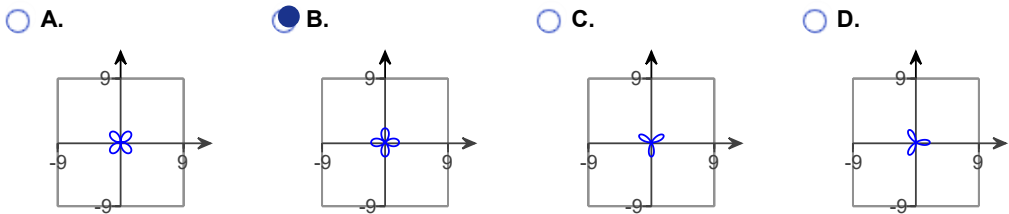
$r = 7$

Choose the correct graph below.



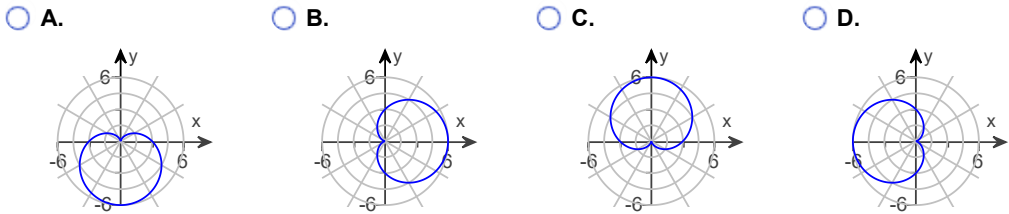
55. Which of the graphs shown below is the graph of the polar equation $r = 2 \cos 2\theta$?

Choose the correct graph below.



56. Give a complete graph of the polar equation $r = 3 + 3 \cos \theta$ and identify the type of polar graph.

Choose the correct graph of $r = 3 + 3 \cos \theta$ below.



The polar graph of $r = 3 + 3 \cos \theta$ is a (1) _____.

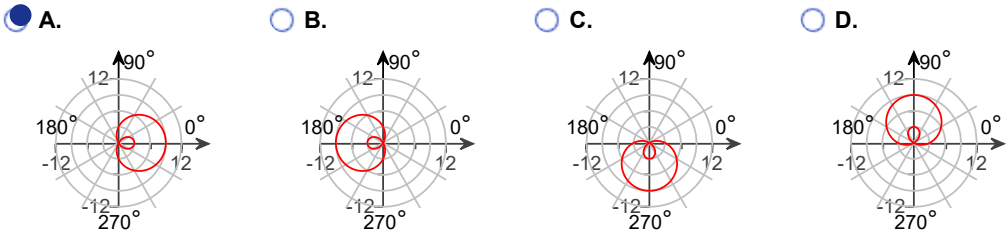
- ☐ circle
- ☐ cardioid
- ☐ lemniscate
- ☐ rose curve
- ☐ limaçon (where $a \neq b$)

55. $n = 2$ even: $2n = 4$ petals, length 2, on the axes

B

57. Give a complete graph of the polar equation $r = 3 + 6 \cos \theta$ and identify the type of polar graph.

Choose the correct graph of $r = 3 + 6 \cos \theta$ below.

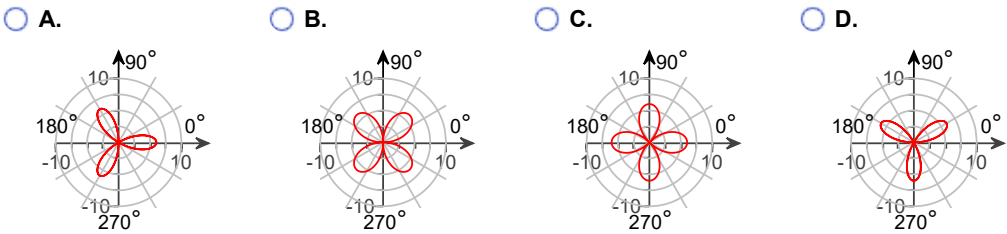


The polar graph of $r = 3 + 6 \cos \theta$ is a (1) _____.

- (1) ☐ lemniscate
- ☐ rose curve
- ☐ cardioid
- ☒ limaçon (where $a \neq b$)
- ☐ circle

58. Give a complete graph of the polar equation $r = 6 \cos 2\theta$ and identify the type of polar graph.

Choose the correct graph of $r = 6 \cos 2\theta$ below.

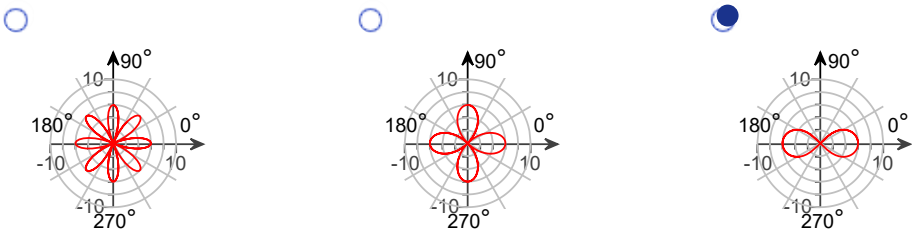


The polar graph of $r = 6 \cos 2\theta$ is a (1) _____.

- (1) ☐ rose curve
- ☐ cardioid
- ☐ lemniscate
- ☐ limaçon (where $a \neq b$)
- ☐ circle

59. Give a complete graph of the polar equation $r^2 = 36 \cos 2\theta$ and identify the type of polar graph.

Choose the correct graph of $r^2 = 36 \cos 2\theta$ below.



The polar graph of $r^2 = 36 \cos 2\theta$ is a (1) _____.

- (1) ☒ lemniscate
- ☐ limaçon (where $a \neq b$)
- ☐ circle
- ☐ cardioid
- ☐ rose curve

57. $\frac{a}{b} = \frac{3}{6} = \frac{1}{2} < 1$: inner loop
 $\theta = 0^\circ$: $r = 9$; $\theta = 180^\circ$: $r = -3$

A limaçon (where $a \neq b$)

59. $r = \pm 6\sqrt{\cos 2\theta}$; $\theta = 0^\circ$: $r = \pm 6$
loops on the x -axis: third graph

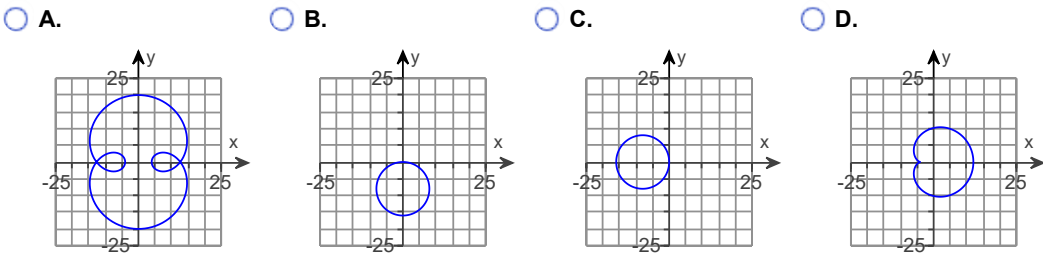
lemniscate

60. Convert the polar equation to rectangular form and identify the graph.

$r = -16 \cos \theta$

The polar equation $r = -16 \cos \theta$ is equivalent to the rectangular equation .
(Type an equation. Type your answer in standard form.)

Identify the graph of the rectangular equation found above. Choose the correct answer below.



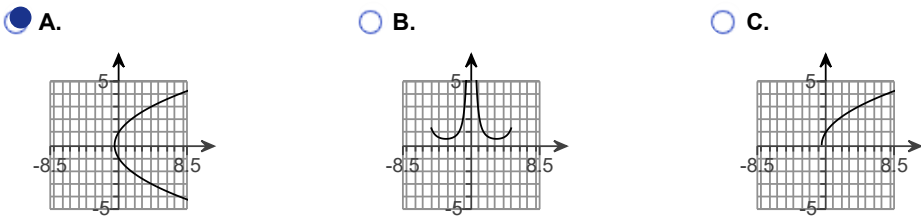
61. For the given equation, find an equivalent equation in rectangular coordinates and graph.

$r = \frac{1}{1 - \cos \theta}$

Choose an equivalent equation below.

- ☐ A. $\sqrt{x^2 + y^2} = x - 1$
- ☐ B. $y = \frac{1}{1 - \cos x}$
- ☐ C. $y^2 = 1$
- ☒ D. $x = \frac{y^2 - 1}{2}$

Choose the correct graph below.



61. $r(1 - \cos \theta) = 1$
 $r - r \cos \theta = 1$
 $r - x = 1 \Rightarrow r = x + 1$
 $x^2 + y^2 = x^2 + 2x + 1$
 $y^2 = 2x + 1$
 $x = \frac{y^2 - 1}{2}$
☐ D vertex $(-\frac{1}{2}, 0)$, opens right: ☒ A

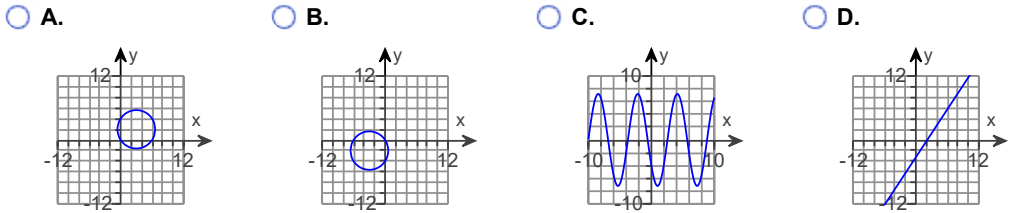
62. Find an equivalent equation in rectangular coordinates and graph the equation.

$r = -6 \cos \theta - 4 \sin \theta$

Choose the rectangular equation that is equivalent to $r = -6 \cos \theta - 4 \sin \theta$.

- ☐ A. $13 = 6x - 4y$
- ☐ B. $y = -6 \sin x - 4 \cos x$
- ☐ C. $(x - 2)^2 + (y - 3)^2 = 13$
- ☐ D. $(x + 3)^2 + (y + 2)^2 = 13$

Choose the correct graph of the equation below.



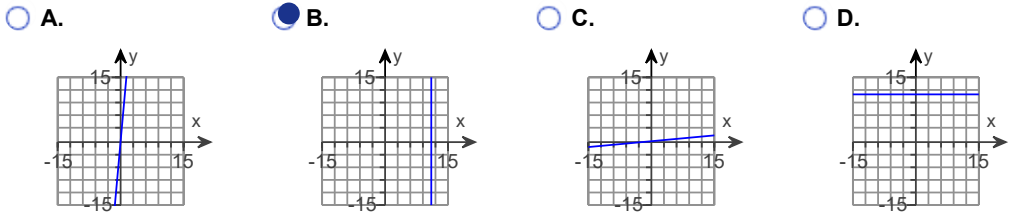
63. Find an equivalent equation in rectangular coordinates and graph the equation.

$r = 11 \sec \theta$

Choose the rectangular equation that is equivalent to $r = 11 \sec \theta$.

- ☐ A. $\sqrt{x^2 + y^2} = 11x$
- ☐ B. $\sqrt{x^2 + y^2} = 11y$
- ☐ C. $y = 11$
- ☒ D. $x = 11$

Choose the correct graph of the equation below.



63. $r = \frac{11}{\cos \theta}$, $r \cos \theta = 11$, $x = 11$:

vertical line at $x = 11$:

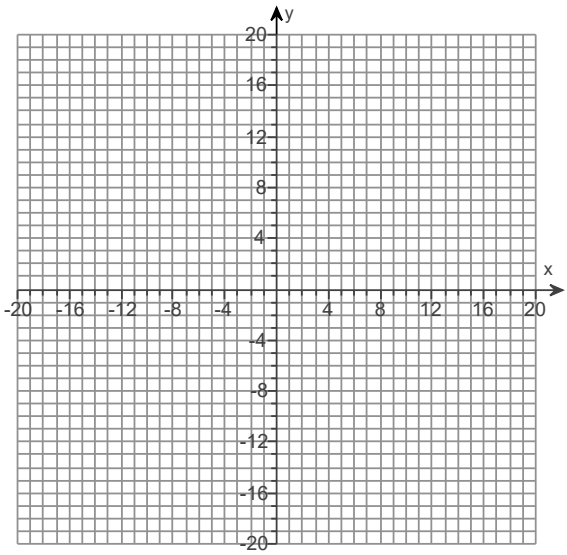
64. Sketch the graph of the following polar equation by transforming it to rectangular coordinates.

$$r = \frac{7}{\cos \theta + \sin \theta}$$

The rectangular equation for $r = \frac{7}{\cos \theta + \sin \theta}$ is .

(Type an equation. Use integers or decimals for any numbers in the equation.)

Use the graphing tool to graph the equation.



65. Find the polar equation of the line that passes through $(3,0^\circ)$ and $(2,90^\circ)$.

The equation is $r = \frac{6}{2 \cos \theta + 3 \sin \theta}$
(Simplify your answer.)

66. Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ .

$r = 24 \sin \theta, r = 8 + 8 \sin \theta \quad 0 \leq \theta < 2\pi$

Choose the correct solutions below.

- ☐ A. $\frac{\pi}{6}, \frac{5\pi}{6}$
- ☐ B. $\left(\frac{\pi}{6}, 12\right), \left(\frac{5\pi}{6}, 12\right)$
- ☐ C. $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
- ☐ D. $\left(12, \frac{\pi}{6}\right), \left(12, \frac{5\pi}{6}\right)$

67. Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ .

$r = 7 + \sin \theta, r = 7 - \cos \theta; 0 \leq \theta < 2\pi$

Choose the correct solutions below.

- ☐ A. $\left(\frac{7 + \sqrt{2}}{2}, \frac{3\pi}{4}\right), \left(\frac{7 - \sqrt{2}}{2}, \frac{7\pi}{4}\right)$
- ☐ B. $\left(\frac{14 - \sqrt{2}}{2}, \frac{3\pi}{4}\right), \left(\frac{14 + \sqrt{2}}{2}, \frac{7\pi}{4}\right)$
- ☐ C. $\left(\frac{14 + \sqrt{2}}{2}, \frac{3\pi}{2}\right), \left(\frac{14 - \sqrt{2}}{2}, \frac{7\pi}{6}\right)$
- ☒ D. $\left(\frac{14 + \sqrt{2}}{2}, \frac{3\pi}{4}\right), \left(\frac{14 - \sqrt{2}}{2}, \frac{7\pi}{4}\right)$

65. $(3,0^\circ) \rightarrow (3,0), (2,90^\circ) \rightarrow (0,2)$

$m = \frac{2-0}{0-3} = -\frac{2}{3}, y = -\frac{2}{3}x + 2$

$3y = -2x + 6, 2x + 3y = 6$

$2r \cos \theta + 3r \sin \theta = 6$

$r = \frac{6}{2 \cos \theta + 3 \sin \theta}$

67. $7 + \sin \theta = 7 - \cos \theta$

$\sin \theta = -\cos \theta \Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

$r\left(\frac{3\pi}{4}\right) = 7 + \frac{\sqrt{2}}{2} = \frac{14 + \sqrt{2}}{2}$

$r\left(\frac{7\pi}{4}\right) = 7 - \frac{\sqrt{2}}{2} = \frac{14 - \sqrt{2}}{2}$

D

1. True

2. True

3. A. The given statement is true because any number can be written in the form of a complex number $a + bi$.

4. B. pure imaginary, C. nonreal complex, D. complex

5. C. Complex, D. Nonreal complex

6. B. Complex, C. Nonreal complex, D. Pure imaginary

7. $i\sqrt{19}$

8. $5i\sqrt{3}$

9. -14

10. $\sqrt{3}$

11. $2i$

12. $-5 + i\sqrt{10}$

13. 3

14. 0

15. $-4 - 28i$

16. $-32 - 126i$

17. 82

18. 25

19. $61i$

20. $\frac{57}{65} - \frac{1}{65}i$

21. $2i$

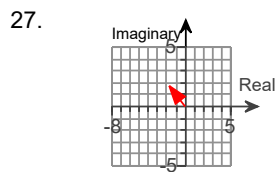
22. $-2i\sqrt{5}, 2i\sqrt{5}$

23. $-\frac{2}{7} + \frac{\sqrt{17}}{7}i, -\frac{2}{7} - \frac{\sqrt{17}}{7}i$

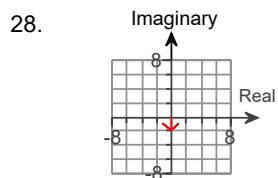
24. $-i$

25. $-i$

26. $-i$



D.



D.

29. B. $9\sqrt{2} + 9i\sqrt{2}$

30. $-\frac{9}{2} + \frac{9\sqrt{3}}{2}i$

31. $10\sqrt{2} - 10i\sqrt{2}$

32. $4\sqrt{2}$

45

45

33. 5

90

90

34. Yes

35. $7\sqrt{2} + 7i\sqrt{2}$

36. $5i$

37. -6

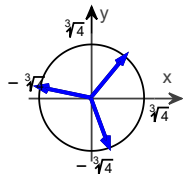
38. $-\frac{\sqrt{2}}{10} + \frac{\sqrt{2}}{10}i$

39. $7i$

40. $-108\sqrt{3} + 108i$

41. $17496 - 17496i$

42. B. $\sqrt[3]{4}(\cos 50^\circ + i \sin 50^\circ), \sqrt[3]{4}(\cos 170^\circ + i \sin 170^\circ), \sqrt[3]{4}(\cos 290^\circ + i \sin 290^\circ)$



B.

43. C. $\{2(\cos 30^\circ + i \sin 30^\circ), 2(\cos 150^\circ + i \sin 150^\circ), 2(\cos 270^\circ + i \sin 270^\circ)\}$

44. C. $\{\sqrt[4]{8} \operatorname{cis} 45^\circ, \sqrt[4]{8} \operatorname{cis} 135^\circ, \sqrt[4]{8} \operatorname{cis} 225^\circ, \sqrt[4]{8} \operatorname{cis} 315^\circ\}$

45. A. $\{4(\cos 20^\circ + i \sin 20^\circ), 4(\cos 140^\circ + i \sin 140^\circ), 4(\cos 260^\circ + i \sin 260^\circ)\}$

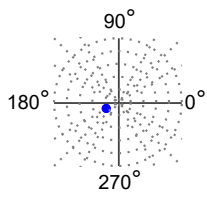
46. B. This number belongs to the Mandelbrot set.

A. This number does not belong to the Mandelbrot set.

B. This number belongs to the Mandelbrot set.

47. II

48.

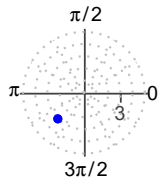


C.

B. $(1, 210^\circ)$, D. $(-1, 390^\circ)$

$$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right)$$

49.



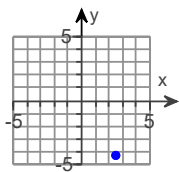
D.

$$\left(3, \frac{13\pi}{4} \right)$$

$$\left(-3, \frac{\pi}{4} \right)$$

$$\left(-\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2} \right)$$

50.

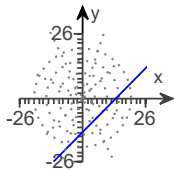


B.

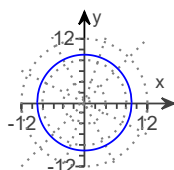
 $(5, 300), (-5, 120)$

51.

$$r = \frac{14}{\cos \theta - \sin \theta}$$

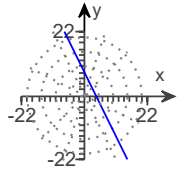


A.

52. $r = 9, r = -9$ 

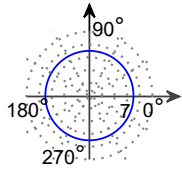
C.

53.
$$r = \frac{8}{2 \cos \theta + \sin \theta}$$



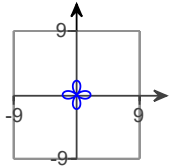
C.

54.



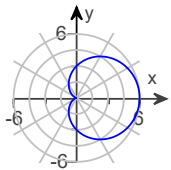
D.

55.



B.

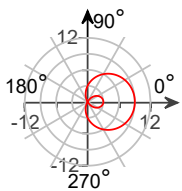
56.



B.

(1) cardioid

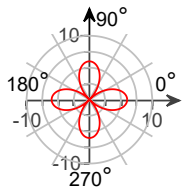
57.



A.

(1) limaçon (where $a \neq b$)

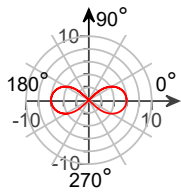
58.



C.

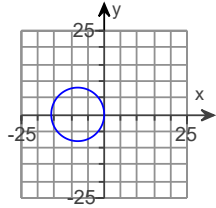
(1) rose curve

59.



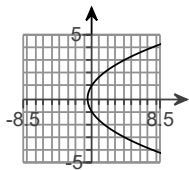
(1) lemniscate

60. $(x+8)^2 + y^2 = 64$



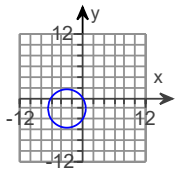
C.

61. D. $x = \frac{y^2 - 1}{2}$



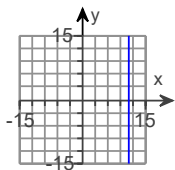
A.

62. D. $(x+3)^2 + (y+2)^2 = 13$



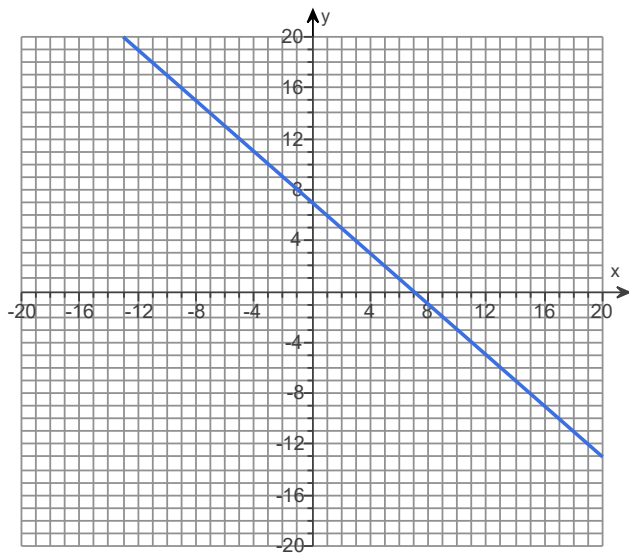
B.

63. D. $x = 11$



B.

64. $x + y = 7$



65.
$$\frac{6}{2 \cos \theta + 3 \sin \theta}$$

66. D. $\left(12, \frac{\pi}{6}\right), \left(12, \frac{5\pi}{6}\right)$

67. D. $\left(\frac{14 + \sqrt{2}}{2}, \frac{3\pi}{4}\right), \left(\frac{14 - \sqrt{2}}{2}, \frac{7\pi}{4}\right)$
